

DD: FUL-03 v1.1 — Service Activation Fulfillment

Developer implementation guide: DD_FUL-03-Service_Activation-DEV_IMPL_GUIDE-v1.1.md.

1. Purpose

This rewritten DD is the developer-facing version of the module contract DD_FUL-03-Service_Activation-v1.1.md.

Verdict: ■ Updated | **Wave:** 1 | **Group I:** Fulfillment | **Version:** v1.1 Orchestrates the technical fulfillment of "make service live" intents against a customer's services. Called synchronously by subscription workflows (ACTIVATE-01, RESUME-01, UPGRADE-01, DOWNGRADE-01, RELOCATION-01, MIGRATION-01) via the framework's `ful-call-activate` worker (for `ACTIVATE_SERVICE` / `RESUME_SERVICE`) or `ful-call-modify` worker (for `MODIFY_SERVICE`). FUL-03 loads the Package's constituent Services (per SIP-01), iterates each Service, resolves the target network node + port via the customer's HomePass `service_management_endpoints` map (per RLM-CFG-01), and dispatches to the Provisioner adapter registered for the Service's `provisionerKey` (per PLM-CFG-01). For `MODIFY_SERVICE`, FUL-03 also performs the per-service-class diff (source composition vs target composition) and dispatches `activate/deactivate/modify` verbs per class. Aggregates outcomes, supports compensation on partial failure, returns a structured outcome. **What changed from v1.0:** v1.0 supported one intent (ACTIVATE) with one endpoint (`POST /api/activation/execute`). v1.1 adopts the FUL-04-style multi-intent envelope: a unified endpoint `POST /api/service-fulfillment/execute` accepting intent $\in \{\text{ACTIVATE_SERVICE, MODIFY_SERVICE, RESUME_SERVICE}\}$. The v1.0 endpoint remains as a backwards-compatible alias for `ACTIVATE_SERVICE` (deprecated; callers migrate to the new endpoint in their next minor). **Companion DDs:** FUL-04 (Restriction Fulfillment — owns `SUSPEND_SERVICE` / `APPLY_RESTRICTION` / `LIFT_RESTRICTION`), FUL-05 (Termination Fulfillment — owns network revocation), FUL-07 (HomePass Lifecycle Fulfillment — owns `CLAIM` / `RELEASE` bookkeeping). FUL-03 stays in its "make service live" lane.

The goal of this rewrite is to make the DD easier for a mid-level developer to implement without losing the detailed source contract.

2. Implementation Scope

This DD should be implemented as part of the owning SOPHIX capability, not as an isolated service unless the deployment architecture explicitly separates that capability.

Implement this DD with these rules:

- Use the owning module APIs/repositories for authoritative writes.
- Use approved internal APIs/client wrappers when reading another module's data.
- Do not query another module's database tables directly from business flow code.
- Treat worker names as logical Camunda external-task topics, not separate deployments.
- One worker application may handle multiple topics, but metrics, retries, and logs must remain visible per topic.
- Emit success events only after the authoritative state commit succeeds.
- Use outbox publishing for Kafka events.

3. Runtime Metadata

Item	Value
Source file	/Users/macbook/Downloads/Sophix_V3_BSS_DDs_MD_2026-05-27/06_fulfillment/DD_FUL-03-Service_Activation-v1.1.md
Version	v1.1
DD type	module contract
BPMN/process keys	Not explicitly defined
Drools packages	Not explicitly defined

4. Runtime Contracts To Implement

4.1 APIs

- GET /homepasses/hp_01HZ_KAREN_017
- GET /homepasses/hp_01HZ_KARN_022
- GET /packages/pkg_01HX_INET_100M_VOIP_RES/versions/pkv_01HX_INET_100M_V3
- GET /services/svc_INET_100M
- POST /api/activation/execute
- POST /api/service-fulfillment/execute
- POST /external-task/{taskId}/complete

4.2 Tables

- fulfillment_ledger
- fulfillment_service_outcome

For each table in this DD, implement the schema with:

- a short table comment explaining what the table is for;
- column comments or migration notes explaining each field;
- constraints and indexes from the DD;
- seed data where the DD provides operator-specific sample rows.

4.3 Worker Topics

- ful-call-activate
- ful-call-claim-homepass
- ful-call-modify
- osr-equipment-bind
- osr-equipment-unbind
- sub-lm-revert-to-prior

Worker-topic guidance:

- A BPMN service task uses the topic.
- A worker application subscribes to the topic.
- The topic handler validates input variables, calls the owning module/API, writes outputs back to Camunda, and records metrics.
- Do not treat the topic string as a shell command or Kubernetes deployment name.

4.4 Events

- SubscriptionUpgradeRejected

Event guidance:

- Write events through the transactional outbox.
- Include operation id, aggregate id, operator code, actor, and correlation data.
- Consumers should be able to rebuild downstream state without querying workflow internals.

5. Developer Implementation Checklist

1. Add or verify database migrations and seed rows.
2. Add or verify config rows for the operator/module.
3. Implement request/command DTOs and validation.
4. Implement idempotency and in-flight operation checks where the DD requires them.
5. Implement Drools fact mapping and package deployment where rules are used.

6. Implement BPMN process, service tasks, gateways, timers, and message catches where the DD is a flow.
7. Implement worker-topic handlers using owning module APIs.
8. Implement compensation paths and manual recovery paths.
9. Emit success/rejected events through the outbox.
10. Add smoke tests for happy path, validation failure, dependency failure, and retry/idempotency.

6. Infrastructure And AWS Notes

Deploy this DD with the existing SOPHIX platform components unless the owning module requires isolation:

Concern	Recommendation
API/runtime	Run inside the owning module deployment or existing workflow API pods.
Workers	Consolidate low-volume topics into shared worker apps; keep per-topic metrics.
Database	Use the owning module PostgreSQL schema and migration pipeline.
Kafka	Use the foundation Kafka/MSK setup and transactional outbox.
Camunda	Deploy BPMN files through the workflow deployment pipeline.
Drools	Deploy KJAR/rule artifacts through the approved rules pipeline.
Secrets	Use the platform secret manager; on AWS prefer Secrets Manager/Parameter Store with IRSA.
Observability	Alert on stuck operations, failed workers, outbox backlog, and external dependency failures.

7. Original DD Detail Preserved

The following section preserves the detailed source contract for implementation and review.

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Companion DDs: FUL-04 (Restriction Fulfillment — owns `SUSPEND_SERVICE` / `APPLY_RESTRICTION` / `LIFT_RESTRICTION`), FUL-05 (Termination Fulfillment — owns network revocation), FUL-07 (HomePass Lifecycle Fulfillment — owns `CLAIM` / `RELEASE` bookkeeping). FUL-03 stays in its "make service live" lane.

Glossary

Term	Meaning
Intent	The verb FUL-03 is executing: <code>ACTIVATE_SERVICE</code> / <code>MODIFY_SERVICE</code> / <code>RESUME_SERVICE</code> . Carried in the request envelope; drives which Provisioner interface method is invoked per Service.
<code>ACTIVATE_SERVICE</code>	Bring a previously inactive Package's Services live at the given HomePass. Used by: <code>ACTIVATE-01</code> (fresh activation), <code>RELOCATION-01</code> (activate at target), <code>MIGRATION-01</code> (activate at target with target Package).

Term	Meaning
MODIFY_SERVICE	Change a subscription's active Service composition. Used by: UPGRADE-01 + DOWNGRADE-01. FUL-03 diffs source composition vs target composition and dispatches per-class verbs (activate new classes, deactivate dropped classes, modify changed classes).
RESUME_SERVICE	Re-activate Services that were previously suspended by FUL-04 (customer-initiated PAUSE or non-payment SUSPEND_NP). Distinct from ACTIVATE_SERVICE: RESUME assumes the network-side state still exists (port still provisioned, ONT still on the OLT) and lifts only the suspension flag. ACTIVATE_SERVICE assumes the service is being newly stood up at the HomePass.
HomePass	A row in RLM-CFG-01's HomePass catalog. Represents the customer's physical install location. Carries the network_path (upstream nodes) and the derived service_management_endpoints map.
service_management_endpoints	A pre-computed JSONB map on the HomePass: { serviceManagementKey → { nodeCode, port } }. Service families (KE seed): data, voice, iptv_multicast, monitoring. Per RLM-CFG-01, derived from network_path at save time.
serviceManagementKey	A field on each Service (PLM-CFG-01). Tells FUL-03 which key to look up in the HomePass's service_management_endpoints map.
Source composition / Target composition	For MODIFY_SERVICE only. Source = the Package composition before the change. Target = the Package composition after. FUL-03 computes the diff (addedServiceClasses, droppedServiceClasses, modifiedServiceClasses) and dispatches per-class.
Per-service-class fan-out	For Packages with multiple service classes (e.g., {INET, VOIP, IPTV}), FUL-03 dispatches one Provisioner call per class, each resolving its own (nodeCode, port) via the Service's serviceManagementKey.
Target node code + port	The pair (nodeCode, port) resolved from homepass.service_management_endpoints[service.serviceManagementKey]. Passed to the Provisioner adapter at fulfillment time.
Provisioner adapter	A code module that translates a FUL-03 verb into vendor commands (e.g., GponResProvisioner.activateService(...) calls Huawei NCE). Registered per provisionerKey per PLM-CFG-01.
Fulfillment correlation ID	Caller-supplied ID used for idempotency and saga correlation. FUL-03 returns the same outcome row on retry with the same ID.

A — Target

1. What this process is about

A subscription workflow (ACTIVATE, RESUME, UPGRADE, DOWNGRADE, RELOCATION, MIGRATION) reaches its fulfillment step. The workflow's BPMN dispatches a worker (ful-call-activate or ful-call-modify) which calls FUL-03's REST endpoint synchronously. FUL-03 then:

1. Loads the subscription's current Package + the Package's Services from SIP-01 + PLM-CFG-01.
2. For MODIFY_SERVICE: also loads the target Package composition (carried in envelope).
3. Loads the HomePass + service_management_endpoints from RLM-CFG-01.
4. For each Service to dispatch:
 - Resolves target via `homepass.service_management_endpoints[service.serviceManagementKey] → {nodeCode, port}`.
 - If serviceManagementKey is null (partner-fulfilled service), passes (null, null) to the adapter.
 - Looks up Provisioner adapter by `service.provisionerKey`.
 - Invokes the verb matching intent:
 - ACTIVATE_SERVICE → `Provisioner.activateService(...)`
 - MODIFY_SERVICE for an added class → `Provisioner.activateService(...)`
 - MODIFY_SERVICE for a dropped class → `Provisioner.deactivateService(...)`
 - MODIFY_SERVICE for a modified class (same class, different profile params) → `Provisioner.modifyService(oldParams, newParams, ...)`

- RESUME_SERVICE → Provisioner.resumeService(...)
- For ACTIVATE_SERVICE calls, also runs per-customer network-side setup specific to the service class — e.g., for IPTV: provision IGMP-snooping on the port, join the customer's chosen multicast groups, register STB serial in IPTV middleware AAA.
- 5. Aggregates per-Service outcomes into a terminal state (FULLY_PROVISIONED / PARTIALLY_PROVISIONED / FAILED_AT_FIRST_SERVICE / FAILED_AND_ROLLED_BACK).
- 6. On PARTIALLY_PROVISIONED or FAILED_AND_ROLLED_BACK: runs compensation by invoking the inverse Provisioner verb on each previously-succeeded Service.
- 7. Returns to the caller synchronously within the 60-second time budget.

The HomePass is not mutated by FUL-03. RLM-CFG-01 is read-only. HomePass-side bookkeeping (counter + binding row) is owned by FUL-07, called by the workflow separately.

2. Business rules

2.1 Entry points

Rule	Description
R-FUL-03-EN-1	FUL-03 v1.1 exposes ONE primary endpoint: POST /api/service-fulfillment/execute. The request envelope carries intent ∈ {ACTIVATE_SERVICE, MODIFY_SERVICE, RESUME_SERVICE}, dispatched per §2.4.
R-FUL-03-EN-2	The envelope MUST carry: operatorCode, subscriptionId, customerId, packageRef, packageVersionId, homepassId, intent, fulfillmentCorrelationId, callerWorkflowKind, callerWorkflowRefId. Missing any required field → 400 MISSING_REQUIRED_FIELD.
R-FUL-03-EN-3	For intent = MODIFY_SERVICE: envelope MUST additionally carry targetPackageRef, targetPackageVersionId, and an equipmentBindings[] array reflecting the post-modification equipment state. Missing → 400.
R-FUL-03-EN-4	For intent = ACTIVATE_SERVICE: envelope MUST carry equipmentBindings[] (passed by the caller from the install WO). Missing → 400.
R-FUL-03-EN-5	For intent = RESUME_SERVICE: no additional fields. The Provisioner is expected to find existing network-side state for the subscription's services.
R-FUL-03-EN-6	Idempotency: same fulfillmentCorrelationId re-submitted within 24h returns the prior ledger row's outcome with HTTP 200. Different fulfillmentCorrelationId for the same subscription within 24h returns 409 CONCURRENT_FULFILLMENT_IN_PROGRESS if a prior fulfillment is still in flight.
R-FUL-03-EN-7	The total time budget for FUL-03 from request to response is 60 seconds. FUL-03 monitors elapsed time; if overall_timeout_ms is reached mid-orchestration, the in-flight Service is allowed to complete (its timeout governs) but no further Services are dispatched. The chain is treated as a partial failure subject to compensation per policy.
R-FUL-03-EN-8	Backwards-compat: the v1.0 endpoint POST /api/activation/execute remains as an alias for intent = ACTIVATE_SERVICE. Marked Deprecated: use /api/service-fulfillment/execute in HTTP headers. Removal planned for v2.0. Existing ACTIVATE-01 callers continue to work without code change.

2.2 Load + resolve

Rule	Description
R-FUL-03-LO-1	FUL-03 loads the current Package via SIP-01 using packageRef + packageVersionId. Pinned version (the caller's snapshot at request time). Reads serviceRefs[]. If the Package is not found → PACKAGE_NOT_FOUND.
R-FUL-03-LO-2	For each Service ref, loads the Service def via PLM-CFG-01. Reads consumption_model, is_addressable, network_profile_shape, service_management_key, provisioner_key, equipment_requirement. Services with consumption_model != CONTINUOUS are skipped (outcome = SKIPPED_NOT_ADDRESSABLE). Services with is_addressable = false are skipped (outcome = SKIPPED_NOT_ADDRESSABLE).

Rule	Description
R-FUL-03-LO-3	For intent = MODIFY_SERVICE: FUL-03 also loads the target Package via SIP-01 using targetPackageRef + targetPackageVersionId. Reads its serviceRefs[] for diff computation.
R-FUL-03-LO-4	FUL-03 loads the HomePass via RLM-CFG-01 (read-only) using homepassId. Reads service_management_endpoints + status_code. If the HomePass does not exist → HOMEPASS_NOT_FOUND. If the HomePass is in a status blocking activation per RLM-CFG-01's catalog rules (NPL, NSV, NSR for ACTIVATE_SERVICE / MODIFY_SERVICE) → HOMEPASS_NOT_READY. For RESUME_SERVICE on a non-ready HomePass, the chain proceeds (Service outcomes will record what the adapters report).
R-FUL-03-LO-5	For each Service whose service_management_key is non-null, FUL-03 resolves the target via homepass.service_management_endpoints[service.service_management_key] → {nodeCode, port}. Missing key → NO_HOMEPASS_ENDPOINT_FOR_KEY (Service-level failure). Null port → HOMEPASS_PORT_NOT_CAPTURED (Service-level failure).
R-FUL-03-LO-6	For each Service with null service_management_key (partner-fulfilled), passes (null, null) to the Provisioner. The adapter sources its target from its own config.

2.3 MODIFY_SERVICE diff computation

Rule	Description
R-FUL-03-MO-1	For intent = MODIFY_SERVICE, FUL-03 computes three sets by comparing source Package composition (serviceRefs[]) to target Package composition (targetPackageRef.serviceRefs[]): (a) addedServices[] — Services in target but not in source; (b) droppedServices[] — Services in source but not in target; (c) modifiedServices[] — Services in both, BUT with different network_profile_shape.params (e.g., same INET class, different speed tier).
R-FUL-03-MO-2	Dispatch per class: (a) addedServices[] → Provisioner.activateService(...) per Service; (b) droppedServices[] → Provisioner.deactivateService(...) per Service; (c) modifiedServices[] → Provisioner.modifyService(oldParams, newParams, ...) per Service.
R-FUL-03-MO-3	Equipment-binding deltas are NOT handled by FUL-03. The caller (UPGRADE-01 / DOWNGRADE-01) is responsible for: (a) ensuring required equipment is bound BEFORE calling FUL-03 (osr-equipment-bind worker), and (b) unbinding obsolete equipment AFTER FUL-03 succeeds (osr-equipment-unbind worker). The envelope's equipmentBindings[] carries the post-modification bindings, which FUL-03 passes to Provisioners but does not validate.
R-FUL-03-MO-4	Dispatch order for MODIFY_SERVICE: (a) added classes first (so customer sees new functionality during the change window even if a dropped class is still up); (b) modified classes next; (c) dropped classes last. The Package owner may declare a different order in Package.modificationOrder; FUL-03 honors it.
R-FUL-03-MO-5	If both source and target Service lists are identical (no diff), FUL-03 returns terminal_state = NO_CHANGE immediately with serviceOutcomes = []. The caller treats this as success. (Edge case; should not normally occur — UPGRADE-01 / DOWNGRADE-01's own validation rejects same-Package requests upstream.)

2.4 Per-service dispatch

Rule	Description
R-FUL-03-DI-1	For intent = ACTIVATE_SERVICE: invokes Provisioner.activateService(equipmentBindings[serviceClass], params, customer, targetNodeCode, targetPort). params is service.networkProfileShape.params. Returns ProvisionerOutcome with success, adapterRefs, failureReason, failureDetail.
R-FUL-03-DI-2	For intent = MODIFY_SERVICE: per §2.3, dispatches activateService / deactivateService / modifyService per the diff result. Per-call returns aggregated into one outcome.
R-FUL-03-DI-3	For intent = RESUME_SERVICE: invokes Provisioner.resumeService(params, customer, targetNodeCode, targetPort). The adapter restores the AAA / network-side state suspended by FUL-04. Returns ProvisionerOutcome.

Rule	Description
R-FUL-03-DI-4	The Provisioner interface (per PLM-CFG-01 v1.0 — pending) declares five methods: activateService, modifyService, deactivateService, resumeService, suspendService. FUL-03 uses the first four; FUL-04 owns suspendService (and applyRestriction / liftRestriction which are FUL-04-only). PLM-CFG-01 v1.0 must declare this full interface before FUL-03 v1.1 + FUL-04 v1.1 ship.
R-FUL-03-DI-5	Per-customer network-side setup specific to a service class is the Provisioner adapter's responsibility, dispatched during activateService or modifyService. Examples: for IPTV multicast → IGMP-snooping enablement on the port + multicast group joins + STB AAA registration in middleware. FUL-03 does not handle these specifics; it dispatches the verb and the adapter knows what to do.

2.5 Compensation

Rule	Description
R-FUL-03-CO-1	Compensation invoked when terminal state is FAILED_AND_ROLLED_BACK. For each previously-successful Service, FUL-03 calls the inverse Provisioner verb: activateService → deactivateService; modifyService → modifyService(oldParams = newParams, newParams = oldParams) (reverse-direction modify); deactivateService → activateService; resumeService → suspendService. Order: reverse of original dispatch.
R-FUL-03-CO-2	If compensation itself fails for any Service: terminal compensation state recorded as COMPENSATION_FAILED in the ledger; emit CompensationFailed Kafka event; manual NOC recovery required. The fulfillment ledger row is the audit anchor.
R-FUL-03-CO-3	A failed MODIFY_SERVICE rollback returns the subscription's network state to its pre-modification configuration. The caller (UPGRADE-01 / DOWNGRADE-01) is responsible for reverting its own state changes (package_ref, equipment bindings, billing).

2.6 Outcome aggregation

Rule	Description
R-FUL-03-OA-1	Terminal state = FULLY_PROVISIONED when all dispatched Services succeed.
R-FUL-03-OA-2	Terminal state = PARTIALLY_PROVISIONED when some Services succeed and some fail BUT the failure mode is policy = TOLERATE_PARTIAL. The caller decides whether to accept partial or revert.
R-FUL-03-OA-3	Terminal state = FAILED_AT_FIRST_SERVICE when the first Service fails AND policy = ABORT_ON_FIRST_FAIL (the default). No compensation needed (no prior successes).
R-FUL-03-OA-4	Terminal state = FAILED_AND_ROLLED_BACK when a Service fails AFTER prior successes AND policy requires rollback. Compensation runs per §2.5.
R-FUL-03-OA-5	Per-Service outcomes are recorded in fulfillment_service_outcome (one row per Service per fulfillment). Carries service_ref, outcome (SUCCEEDED / FAILED / SKIPPED_NOT_ADDRESSABLE / COMPENSATED / COMPENSATION_FAILED), adapter_refs (vendor-system IDs), failure_reason, failure_detail, target_node_code, target_port, duration_ms.

2.7 Events

Rule	Description
R-FUL-03-EV-1	On dispatch start, emit ServiceFulfillmentStarted to sophix.fulfillment.service.started. Carries fulfillmentId, subscriptionId, customerId, operatorCode, intent, packageRef, homepassId, startedAt.
R-FUL-03-EV-2	On successful terminal state (FULLY_PROVISIONED / PARTIALLY_PROVISIONED / NO_CHANGE), emit ServiceFulfillmentCompleted to sophix.fulfillment.service.completed. Carries fulfillmentId, subscriptionId, customerId, operatorCode, intent, terminalState, serviceOutcomes[], completedAt.
R-FUL-03-EV-3	On failed terminal state, emit ServiceFulfillmentFailed to sophix.fulfillment.service.failed. Carries fulfillmentId, subscriptionId, customerId, operatorCode, intent, terminalState (FAILED_AT_FIRST_SERVICE / FAILED_AND_ROLLED_BACK), serviceOutcomes[], failureReason, failedAt.

Rule	Description
R-FUL-03-EV-4	On compensation failure, emit CompensationFailed to <code>sophix.fulfillment.service.compensationfailed</code> . Carries <code>fulfillmentId</code> , <code>subscriptionId</code> , <code>failedServiceRefs[]</code> , <code>failureReason</code> . NOC alerting consumer.
R-FUL-03-EV-5	All events carry <code>fulfillmentCorrelationId</code> for cross-system correlation with billing / SUB-LM events.

3. Schemas

3.1 fulfillment_ledger

One row per fulfillment attempt. Append-only.

```
CREATE TABLE fulfillment_ledger (
  id TEXT PRIMARY KEY,
  operator_code TEXT NOT NULL,
  subscription_id TEXT NOT NULL,
  customer_id TEXT NOT NULL,
  package_ref TEXT NOT NULL,
  package_version_id TEXT NOT NULL,
  homepass_id TEXT NOT NULL,
  intent TEXT NOT NULL,
  target_package_ref TEXT NULL,
  target_package_version_id TEXT NULL,
  fulfillment_correlation_id TEXT NOT NULL,
  caller_workflow_kind TEXT NOT NULL,
  caller_workflow_ref_id TEXT NOT NULL,
  terminal_state TEXT NULL,
  services_dispatched_count INTEGER NOT NULL DEFAULT 0,
  services_succeeded_count INTEGER NOT NULL DEFAULT 0,
  services_failed_count INTEGER NOT NULL DEFAULT 0,
  services_skipped_count INTEGER NOT NULL DEFAULT 0,
  failure_reason TEXT NULL,
  duration_ms INTEGER NULL,
  started_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  completed_at TIMESTAMPTZ NULL,
  UNIQUE (subscription_id, fulfillment_correlation_id),
  CHECK (intent IN ('ACTIVATE_SERVICE', 'MODIFY_SERVICE', 'RESUME_SERVICE')),
  CHECK (terminal_state IN ('FULLY_PROVISIONED', 'PARTIALLY_PROVISIONED', 'FAILED_AT_FIRST_SERVICE', 'FAILED_AND_ROLLED_BACK')));
```

Column	Notes
id	ULID. Example: flb_01HZ_KAMAU_ACTIVATE
operator_code	WIK / WUG / WTZ / YASSN
intent	ACTIVATE_SERVICE for fresh activate at HomePass; MODIFY_SERVICE for upgrade/downgrade composition diff; RESUME_SERVICE for post-pause resume
target_package_ref	Set only for intent = MODIFY_SERVICE. Otherwise NULL.
caller_workflow_kind	SUB_WF_ACTIVATE / SUB_WF_RESUME / SUB_WF_UPGRADE / SUB_WF_DOWNGRADE / SUB_WF_RELOCATION / SUB_WF_MIGRATION
terminal_state	NULL while in-flight. Set on completion. NO_CHANGE only valid for MODIFY_SERVICE with empty diff.
failure_reason	Free text. Examples: ADAPTER_TIMEOUT, MISSING_REQUIRED_EQUIPMENT, NO_HOMEPASS_ENDPOINT_FOR_KEY, HOMEPASS_PORT_NOT_CAPTURED, PACKAGE_NOT_FOUND, HOMEPASS_NOT_FOUND, HOMEPASS_NOT_READY

Sample data:

id	subscription_id	intent	package_ref	target_package_ref	terminal_state	services_dispatched_count	services_succeeded_count	duration_ms
flb_01HZ_KAMAU_ACTIVATE	sub_01HZ_KAMAU_FIBER	ACTIVATE_SERVICE	pkg_01HX_INET_100M_VOIP_RES	NULL	FULLY_PROVISIONED	2	2	4250
flb_01HZ_OTIENO_MIGRATE	sub_01HZ_OTIENO_FIBER	ACTIVATE_SERVICE	pkg_01HX_INET_VOIP_IPTV_RES	NULL	FULLY_PROVISIONED	3	3	7180
flb_01HZ_NJERI_UPGRADE	sub_01HZ_NJERI_FIBER	MODIFY_SERVICE	pkg_01HX_INET_100M_VOIP_RES	pkg_01HX_INET_VOIP_IPTV_RES	FULLY_PROVISIONED	1	1	3500

id	subscription_id	intent	package_ref	target_package_ref	terminal_state	services_dispatched_count	services_succeeded_count	duration_ms
flb_01HZ_KIBE_RESUME	sub_01HZ_KIBE_FIBER	RESUME_SERVICE	pkg_01HX_INET_50M	NULL	FULLY_PROVISIONED	1	1	1800
flb_01HZ_WANGARI_DOWNGRADE	sub_01HZ_WANGARI_FIBER	MODIFY_SERVICE	pkg_01HX_INET_VOIP_IPTV_RES	pkg_01HX_INET_100M_VOIP_RES	FULLY_PROVISIONED	1	1	2900

(Row 3: Njeri's upgrade from double-play to triple-play — added IPTV class. Row 5: Wangari's downgrade from triple to double-play — dropped IPTV class.)

3.2 fulfillment_service_outcome

One row per Service per fulfillment. Append-only.

```
CREATE TABLE fulfillment_service_outcome (
  id TEXT PRIMARY KEY,
  fulfillment_id TEXT NOT NULL REFERENCES fulfillment_ledger(id),
  service_ref TEXT NOT NULL,
  service_management_key TEXT NULL,
  provisioner_key TEXT NOT NULL,
  target_node_code TEXT NULL,
  target_port TEXT NULL,
  dispatched_verb TEXT NOT NULL,
  outcome TEXT NOT NULL,
  adapter_refs JSONB NULL,
  failure_reason TEXT NULL,
  failure_detail TEXT NULL,
  compensation_status TEXT NULL,
  compensation_failure_reason TEXT NULL,
  duration_ms INTEGER NULL,
  dispatched_at TIMESTAMPTZ NOT NULL DEFAULT NOW(),
  completed_at TIMESTAMPTZ NULL,
  CHECK (dispatched_verb IN ('activateService', 'modifyService', 'deactivateService', 'resumeService')),
  CHECK (outcome IN ('SUCCEEDED', 'FAILED', 'SKIPPED_NOT_ADDRESSABLE', 'COMPENSATED', 'COMPENSATION_FAILED')),
  CHECK (compensation_status IN ('NOT_NEEDED', 'COMPENSATED', 'COMPENSATION_FAILED', NULL))
);
```

Column	Notes
dispatched_verb	The Provisioner method actually invoked: activateService / modifyService / deactivateService / resumeService. May differ from envelope intent (e.g., MODIFY_SERVICE with dropped class dispatches deactivateService).
target_node_code	Resolved from homepass.service_management_endpoints[key]. Example: OLT-NRB-WTL-01 for a Nairobi OLT. NULL for partner-fulfilled.
adapter_refs	JSONB carrying vendor-system IDs returned by the adapter. Example: {"huaweiCommandId":"cmd_123","provisioningProfileId":"prf_456"}.
outcome	SUCCEEDED / FAILED / SKIPPED_NOT_ADDRESSABLE (non-addressable Service per PLM-CFG-01) / COMPENSATED (was succeeded, then rolled back) / COMPENSATION_FAILED

Sample (for flb_01HZ_NJERI_UPGRADE — upgrade from double-play to triple-play, only IPTV added):

id	fulfillment_id	service_ref	service_management_key	dispatched_verb	outcome	target_node_code	target_port	duration_ms
fso_01HZ_NJERI_IPTV	flb_01HZ_NJERI_UPGRADE	svc_IPTV_BASIS	iptv_multicast	activateService	SUCCEEDED	OLT-NRB-WTL-01	0/2/03	3500

(Only one outcome row — the diff produced one added Service, IPTV. INET and VOIP were unchanged and not dispatched.)

4. Worked examples

Four concrete timelines covering each intent and one failure path with compensation.

4.1 ACTIVATE_SERVICE — fresh activation at install completion

Context: Kamau just completed his install at Karen. His install WO wo_01HZ_KAREN_017_INSTALL finished at 14:32:18 with equipmentBindings=[{eqp_ONT_42, INET}, {eqp_ATA_19, VOIP}]. SUB-WF-ACTIVATE-01's BPMN advances to its callFulfillment step. Kamau's Package is pkg_INET_100M_VOIP_RES (double-play); HomePass

hp_01HZ_KAREN_017 is GPON, service_management_endpoints = {data: {OLT-NRB-KAREN-02, 0/4/07}, voice: {VOIPSWITCH-NRB-01, 5012}}.

14:32:20 — ful-call-activate worker invokes FUL-03. Worker fetches and locks the next external task on topic ful-call-activate, then POSTs to /api/service-fulfillment/execute:

```
POST /api/service-fulfillment/execute
{
  "operatorCode":          "WIK",
  "subscriptionId":        "sub_01HZ_KAMAU_FIBER",
  "customerId":            "cust_01HZ_KAMAU",
  "packageRef":            "pkg_01HX_INET_100M_VOIP_RES",
  "packageVersionId":      "pkv_01HX_INET_100M_V3",
  "homepassId":            "hp_01HZ_KAREN_017",
  "intent":                "ACTIVATE_SERVICE",
  "equipmentBindings":     [
    { "equipmentId": "eqp_ONT_42", "serviceClassId": "INET" },
    { "equipmentId": "eqp_ATA_19", "serviceClassId": "VOIP" }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_KAMAU_ACTIVATE:activate",
  "callerWorkflowKind":      "SUB_WF_ACTIVATE",
  "callerWorkflowRefId":     "subop_01HZ_KAMAU_ACTIVATE"
}
```

14:32:20.1 — Idempotency + ledger init. FUL-03 checks fulfillment_ledger for (subscription_id, fulfillment_correlation_id) = (sub_01HZ_KAMAU_FIBER, subop_01HZ_KAMAU_ACTIVATE:activate). No prior row — inserts a new row flb_01HZ_KAMAU_ACTIVATE with terminal_state = NULL, started_at = NOW.

14:32:20.2 — Load. FUL-03 calls SIP-01 GET

/packages/pkg_01HX_INET_100M_VOIP_RES/versions/pkv_01HX_INET_100M_V3. Returns serviceRefs = [svc_INET_100M, svc_VOIP_STD]. FUL-03 calls PLM-CFG-01 GET /services/svc_INET_100M → {service_management_key: "data", provisioner_key: "gpon-res", network_profile_shape: {speedDownMbps: 100, speedUpMbps: 20, ...}, is_addressable: true, consumption_model: "CONTINUOUS"}. Same for svc_VOIP_STD → {service_management_key: "voice", provisioner_key: "voip-sip", network_profile_shape: {codec: "G.711a", ...}}.

14:32:20.3 — HomePass load. FUL-03 calls RLM-CFG-01 GET /homepasses/hp_01HZ_KAREN_017. Returns service_management_endpoints = {data: {nodeCode: "OLT-NRB-KAREN-02", port: "0/4/07"}, voice: {nodeCode: "VOIPSWITCH-NRB-01", port: "5012"}}, status_code = "RFS". Status check: RFS is sellable + active → proceed.

14:32:20.4 — Dispatch INET. FUL-03 resolves target via service_management_endpoints["data"] → (OLT-NRB-KAREN-02, 0/4/07). Looks up Provisioner adapter for provisioner_key = "gpon-res" → GponResProvisioner bean. Invokes:

```
gponResProvisioner.activateService(
  equipmentBinding: {eqp_ONT_42, INET},
  params: {speedDownMbps: 100, speedUpMbps: 20, ...},
  customer: {customerId: cust_01HZ_KAMAU, ...},
  targetNodeCode: "OLT-NRB-KAREN-02",
  targetPort: "0/4/07"
)
```

The adapter calls Huawei NCE: provisions ONT serial eqp_ONT_42 on port 0/4/07, pushes the 100Mbps profile, marks port IN_SERVICE. Returns ProvisionerOutcome{success: true, adapterRefs: {huaweiCommandId: "cmd_8801", provisioningProfileId: "prf_100m_v3"}} after 1.2 seconds.

FUL-03 inserts row fso_01HZ_KAMAU_INET into fulfillment_service_outcome with outcome=SUCCEEDED, duration_ms=1200.

14:32:21.7 — Dispatch VOIP. Resolves target via service_management_endpoints["voice"] → (VOIPSWITCH-NRB-01, 5012). Invokes voipSipProvisioner.activateService(...). Adapter calls the SIP switch: registers the ATA eqp_ATA_19 at port 5012, provisions Kamau's DID, applies codec policy. Returns {success: true, adapterRefs: {sipRegistrationId: "sip_4421"}} after 1.5 seconds.

FUL-03 inserts row fso_01HZ_KAMAU_VOIP into fulfillment_service_outcome.

14:32:23.2 — Outcome aggregation. All 2 services succeeded. Terminal state FULLY_PROVISIONED. FUL-03 updates ledger row: terminal_state=FULLY_PROVISIONED, services_dispatched_count=2, services_succeeded_count=2, completed_at=NOW, duration_ms=2200.

14:32:23.3 — Emit Kafka. FUL-03 publishes ServiceFulfillmentCompleted to sophix.fulfillment.service.completed via outbox. Returns 200 to caller with the structured response (per §5.1).

14:32:23.4 — Worker completes task. External task worker calls Camunda POST /external-task/{taskId}/complete with the FUL-03 response. BPMN advances; the saga's next step is ful-call-claim-homepass (FUL-07 CLAIM at the same HomePass).

Total elapsed time: 3.4 seconds. Clean happy path.

4.2 MODIFY_SERVICE — upgrade adding a service class

Context: Njeri is on pkg_INET_100M_VOIP_RES (double-play) at HomePass hp_01HZ_KARN_022 and upgrading to pkg_INET_VOIP_IPTV_RES (triple-play). SUB-WF-UPGRADE-01 has already: (a) validated the upgrade, (b) called BIL-01 with UPGRADE_INTENT and got back proration delta, (c) collected payment, (d) called OSR-INSTANCE-01 to bind the new STB eqp_STB_NEW_88 for the IPTV class (a tech delivered it earlier that morning). The BPMN now reaches the ful-call-modify step.

10:35:38 — ful-call-modify worker invokes FUL-03.

```
POST /api/service-fulfillment/execute
{
  "operatorCode":      "WIK",
  "subscriptionId":    "sub_01HZ_NJERI_FIBER",
  "customerId":        "cust_01HZ_NJERI",
  "packageRef":        "pkg_01HX_INET_100M_VOIP_RES",
  "packageVersionId":  "pkv_01HX_INET_100M_V3",
  "targetPackageRef":  "pkg_01HX_INET_VOIP_IPTV_RES",
  "targetPackageVersionId": "pkv_01HX_INET_VOIP_IPTV_V1",
  "homepassId":        "hp_01HZ_KARN_022",
  "intent":            "MODIFY_SERVICE",
  "equipmentBindings": [
    { "equipmentId": "eqp_ONT_99", "serviceClassId": "INET" },
    { "equipmentId": "eqp_ATA_44", "serviceClassId": "VOIP" },
    { "equipmentId": "eqp_STB_NEW_88", "serviceClassId": "IPTV" }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_NJERI_UPGRADE:modify",
  "callerWorkflowKind":      "SUB_WF_UPGRADE",
  "callerWorkflowRefId":     "subop_01HZ_NJERI_UPGRADE"
}
```

10:35:38.1 — Load source + target. FUL-03 loads pkg_01HX_INET_100M_VOIP_RES → serviceRefs=[svc_INET_100M, svc_VOIP_STD]. Loads pkg_01HX_INET_VOIP_IPTV_RES → serviceRefs=[svc_INET_100M, svc_VOIP_STD, svc_IPTV_BASIC].

10:35:38.2 — Diff computation. FUL-03 compares:

- addedServices = target - source = [svc_IPTV_BASIC]
- droppedServices = source - target = []
- modifiedServices = [] (svc_INET_100M and svc_VOIP_STD are in both, same network_profile_shape.params)

Single-Service diff. The INET speed didn't change (still 100Mbps), the VOIP codec didn't change — those services keep running untouched. Only IPTV needs adding.

10:35:38.3 — HomePass load. FUL-03 calls RLM-CFG-01 GET /homepasses/hp_01HZ_KARN_022. Returns service_management_endpoints = {data: {OLT-NRB-WTL-01, 0/2/03}, voice: {VOIPSWITCH-NRB-01, 5099}, iptv_multicast: {OLT-NRB-WTL-01, 0/2/03}}. (Note: IPTV multicast resolves to the SAME OLT port as data — multicast traffic flows over the same fiber as the customer's data, distinguished by IGMP-snooping enabled on the port.)

10:35:38.4 — Dispatch added: IPTV activate. Looks up svc_IPTV_BASIC → {service_management_key: "iptv_multicast", provisioner_key: "gpon-res", network_profile_shape: {multicastGroups: ["epg_zuku_basic"], stbAaaProfile: "basic-tier", ...}}. Resolves target via service_management_endpoints["iptv_multicast"] → (OLT-NRB-WTL-01, 0/2/03).

Invokes gponResProvisioner.activateService(equipmentBinding: {eqp_STB_NEW_88, IPTV}, params: {multicastGroups: ["epg_zuku_basic"], stbAaaProfile: "basic-tier", ...}). The adapter:

1. Configures IGMP-snooping on OLT port 0/2/03 (already enabled? a flag check on the port — if already enabled, no-op; else enable and reconfigure).
2. Joins the customer's chosen multicast group(s) for the basic tier.

- Registers the STB serial eqp_STB_NEW_88 in the IPTV middleware AAA (call to Verimatrix middleware via HTTPS): grants the STB access to the channel package, pushes EPG entitlements.
- Returns {success: true, adapterRefs: {middlewareAaaId: "aaa_661", multicastGroupSubscriptionId: "mc_sub_770"}} after 3.5 seconds.

FUL-03 inserts fso_01HZ_NJERI_IPTV with dispatched_verb=activateService, outcome=SUCCEEDED, duration_ms=3500.

10:35:42.1 — No more diff entries. No dropped services, no modified services. The remaining INET and VOIP services keep running on the existing provisioning — FUL-03 explicitly does NOT touch them. The customer's broadband stays up throughout the upgrade.

10:35:42.2 — Outcome aggregation. 1 service dispatched, 1 succeeded. Terminal state FULLY_PROVISIONED. Ledger updated. ServiceFulfillmentCompleted emitted.

Response to caller (per §5.2): terminalState: FULLY_PROVISIONED, diffSummary: {addedServices: [svc_IPTV_BASIC], droppedServices: [], modifiedServices: []}, serviceOutcomes: [...]. UPGRADE-01's BPMN advances to commit master fields (writes package_ref = pkg_01HX_INET_VOIP_IPTV_RES).

Total elapsed time: 4.1 seconds. Existing INET + VOIP untouched.

4.3 RESUME_SERVICE — post-pause restoration

Context: Kibe paused his service on May 1 because he was traveling. FUL-04 had run SUSPEND_SERVICE at that time — the AAA disabled his SIP registration and rate-limited his OLT port. On May 15 he returns and calls CC to resume. SUB-WF-RESUME-01 validates (subscription in SUSPENDED, not non-payment) and reaches ful-call-activate with intent=RESUME_SERVICE.

16:42:10 — ful-call-activate worker invokes FUL-03 with RESUME_SERVICE intent.

```
POST /api/service-fulfillment/execute
{
  "operatorCode":          "WIK",
  "subscriptionId":        "sub_01HZ_KIBE_FIBER",
  "customerId":            "cust_01HZ_KIBE",
  "packageRef":            "pkg_01HX_INET_50M",
  "packageVersionId":      "pkv_01HX_INET_50M_V2",
  "homepassId":           "hp_01HZ_NGNG_044",
  "intent":                "RESUME_SERVICE",
  "fulfillmentCorrelationId": "subop_01HZ_KIBE_RESUME:resume",
  "callerWorkflowKind":     "SUB_WF_RESUME",
  "callerWorkflowRefId":    "subop_01HZ_KIBE_RESUME"
}
```

(No equipmentBindings[] needed — equipment is already bound from before the pause. RESUME assumes network state still exists.)

16:42:10.1 — Load + HomePass. Same pattern as §4.1. Package pkg_INET_50M has one Service svc_INET_50M. HomePass hp_01HZ_NGNG_044 is GPON, port 0/4/12 on OLT-NRB-NGNG-02. Status ACT (active subscriber on the HomePass — Kibe is the active subscriber even though paused).

16:42:10.2 — Dispatch: resumeService. Resolves service_management_endpoints["data"] → (OLT-NRB-NGNG-02, 0/4/12). Looks up GponResProvisioner bean. Invokes:

```
gponResProvisioner.resumeService(
  params: {speedDownMbps: 50, speedUpMbps: 10, ...},
  customer: {customerId: cust_01HZ_KIBE, ...},
  targetNodeCode: "OLT-NRB-NGNG-02",
  targetPort: "0/4/12"
)
```

The adapter: lifts the AAA suspension flag (re-enables Kibe's subscriber profile in the RADIUS server), removes the rate-zero policy on OLT port 0/4/12. The ONT was never deauthenticated during pause (per pause policy) so it doesn't need to re-register. Customer's connection resumes within ~30 seconds. Adapter returns {success: true, adapterRefs: {huaweiCommandId: "cmd_2207", aaaSessionId: "aaa_sess_8809"}} after 1.8 seconds.

FUL-03 inserts fso_01HZ_KIBE_INET with dispatched_verb=resumeService, outcome=SUCCEEDED, duration_ms=1800.

16:42:11.9 — Outcome aggregation. Terminal state FULLY_PROVISIONED. RESUME-01's BPMN advances to commit state ACTIVE.

Distinction from ACTIVATE_SERVICE: ACTIVATE_SERVICE assumes the network-side state needs to be stood up from scratch (provision the ONT, push the speed profile, register the SIP endpoint). For a resume, that state already exists — the pause only suspended access flags at the AAA layer + applied a rate-zero policy at the OLT. RESUME_SERVICE tells the adapter "lift the suspension and rate-zero; the rest of the configuration is already in place." Distinct verbs let the adapter make this distinction without FUL-03 needing to know the network-side state.

4.4 Failure with compensation — adapter timeout mid-flow

Context: Same setup as §4.2 (Njeri upgrade to triple-play), but the Verimatrix middleware is unreachable today. The IPTV adapter call times out at 25 seconds.

10:35:38 — ful-call-modify worker invokes FUL-03 (same request body as §4.2).

10:35:38.1 — Load + diff. Same as §4.2: 1 added service (svc_IPTV_BASIC), 0 dropped, 0 modified.

10:35:38.4 — Dispatch IPTV activate. `gponResProvisioner.activateService(...)` — the adapter:

1. IGMP-snooping enable on port 0/2/03 → success (1.2s).
2. Multicast group join → success (0.8s).
3. STB AAA registration in Verimatrix → **HTTP request hangs**. After 25 seconds, the adapter's HTTP client times out.

Adapter returns `{success: false, failureReason: "ADAPTER_TIMEOUT", failureDetail: "Verimatrix middleware timeout after 25s – connection to https://verimatrix.wananchi-internal/api/aaa/register hung at TCP handshake"}` after 27 seconds total.

FUL-03 inserts `fso_01HZ_NJERI_IPTV_FAIL` with `outcome=FAILED`, `dispatched_verb=activateService`, `failure_reason=ADAPTER_TIMEOUT`, `duration_ms=27000`.

10:36:05.5 — Outcome aggregation: failure. First service dispatched, first service failed. Per `policy=ABORT_ON_FIRST_FAIL`, terminal state = `FAILED_AT_FIRST_SERVICE` (not `FAILED_AND_ROLLED_BACK` — no prior successes to roll back).

But wait: the adapter DID partially apply state — IGMP-snooping is enabled on the port and the multicast group is joined. Even though the AAA registration failed, the network plane has partial state. The adapter's contract requires it to self-roll-back its own intermediate state on failure (the adapter should call its own teardown on the IGMP enable + multicast join when it knows the AAA registration won't complete). The adapter does this internally; FUL-03 doesn't see it.

If for some reason the adapter cannot self-roll-back (e.g., the OLT is also unreachable now), the adapter returns `failure_reason=ADAPTER_ERROR` with `compensationState=ADAPTER_INTERNAL_ROLLBACK_FAILED`. FUL-03 records this and emits `CompensationFailed` to NOC.

10:36:05.6 — Emit Kafka. `ServiceFulfillmentFailed` payload includes `terminalState=FAILED_AT_FIRST_SERVICE`, `failureReason=ADAPTER_TIMEOUT`, `serviceOutcomes=[{serviceRef: svc_IPTV_BASIC, outcome: FAILED, ...}]`. Returns 200 to caller (FUL-03 doesn't return non-200 — the BPMN catches the failure via the response body's `terminalState`).

10:36:05.7 — UPGRADE-01 BPMN compensation. SUB-WF-UPGRADE-01's BPMN sees `terminalState != FULLY_PROVISIONED` and enters its own compensation: (a) `osr-equipment-unbind` for the STB (it's no longer in use); (b) `sub-lm-revert-to-prior` flips back to ACTIVE on the source Package; (c) BIL-01 `UPGRADE_FAILED_REVERSAL_INTENT` to refund the proration delta; (d) emit `SubscriptionUpgradeRejected` with `rejectionReason=FULFILLMENT_FAILED`, `rejectionDetail=ADAPTER_TIMEOUT` (IPTV middleware).

Failure with prior successes — alternative path: Imagine a more complex case where Njeri's upgrade also changed her INET speed (50Mbps → 100Mbps) AND added IPTV. The diff would be `modifiedServices=[svc_INET_100M]`, `addedServices=[svc_IPTV_BASIC]`. Per dispatch order (§2.3 R-MO-4), added classes first → IPTV first. If IPTV failed, no rollback needed (no prior successes). But if the order were modified first: INET modify succeeds (1.5s), then IPTV fails (27s). FUL-03 enters compensation: invokes `gponResProvisioner.modifyService(oldParams=newINET, newParams=oldINET, ...)` to reverse the INET speed change. If that succeeds, INET back to 50Mbps and terminal state = `FAILED_AND_ROLLED_BACK`. If the rollback also fails, terminal state = `COMPENSATION_FAILED` and NOC alert fires.

This is why FUL-03's dispatch order matters: added-classes-first (the default) minimizes the rollback surface area in failure cases.

5. REST API

5.1 Activate service at HomePass

POST /api/service-fulfillment/execute

Called by ACTIVATE-01's ful-call-activate worker. Also by RELOCATION-01 + MIGRATION-01 at-target step.

POST /api/service-fulfillment/execute HTTP/1.1
Content-Type: application/json

```
{
  "operatorCode":          "WIK",
  "subscriptionId":        "sub_01HZ_OTIENO_FIBER",
  "customerId":            "cust_01HZ_OTIENO",
  "packageRef":            "pkg_01HX_INET_VOIP_IPTV_RES",
  "packageVersionId":      "pkv_01HX_INET_VOIP_IPTV_V1",
  "homepassId":            "hp_01HZ_LAVI_073",
  "intent":                "ACTIVATE_SERVICE",
  "equipmentBindings": [
    { "equipmentId": "eqp_GPON_ONT_900", "serviceClassId": "INET" },
    { "equipmentId": "eqp_GPON_ATA_510", "serviceClassId": "VOIP" },
    { "equipmentId": "eqp_STB_NEW_77", "serviceClassId": "IPTV" }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_OTIENO_MIGRATE:activate-target",
  "callerWorkflowKind":      "SUB_WF_MIGRATION",
  "callerWorkflowRefId":     "subop_01HZ_OTIENO_MIGRATE"
}

→ 200 OK
{
  "fulfillmentId":          "flb_01HZ_OTIENO_MIGRATE_ACTIVATE",
  "terminalState":          "FULLY_PROVISIONED",
  "serviceOutcomes": [
    { "serviceRef": "svc_INET_100M", "outcome": "SUCCEEDED", "targetNodeCode": "OLT-NRB-LAV-01", "targetPort": "0/2/03", "adapterRef": "adp_01HZ_LAVI_073" },
    { "serviceRef": "svc_VOIP_STD", "outcome": "SUCCEEDED", "targetNodeCode": "VOIPSWITCH-NRB-01", "targetPort": "5103", "adapterRef": "adp_01HZ_LAVI_073" },
    { "serviceRef": "svc_IPTV_BASIC", "outcome": "SUCCEEDED", "targetNodeCode": "OLT-NRB-LAV-01", "targetPort": "0/2/03", "adapterRef": "adp_01HZ_LAVI_073" }
  ],
  "durationMs":              4800,
  "completedAt":             "2026-06-01T11:08:10+03:00"
}
```

5.2 Modify service (composition change)

Called by UPGRADE-01 / DOWNGRADE-01's ful-call-modify worker.

POST /api/service-fulfillment/execute HTTP/1.1
Content-Type: application/json

```
{
  "operatorCode":          "WIK",
  "subscriptionId":        "sub_01HZ_NJERI_FIBER",
  "customerId":            "cust_01HZ_NJERI",
  "packageRef":            "pkg_01HX_INET_100M_VOIP_RES",
  "packageVersionId":      "pkv_01HX_INET_100M_V3",
  "targetPackageRef":      "pkg_01HX_INET_VOIP_IPTV_RES",
  "targetPackageVersionId": "pkv_01HX_INET_VOIP_IPTV_V1",
  "homepassId":            "hp_01HZ_KARN_022",
  "intent":                "MODIFY_SERVICE",
  "equipmentBindings": [
    { "equipmentId": "eqp_ONT_99", "serviceClassId": "INET" },
    { "equipmentId": "eqp_ATA_44", "serviceClassId": "VOIP" },
    { "equipmentId": "eqp_STB_NEW_88", "serviceClassId": "IPTV" }
  ],
  "fulfillmentCorrelationId": "subop_01HZ_NJERI_UPGRADE:modify",
  "callerWorkflowKind":      "SUB_WF_UPGRADE",
  "callerWorkflowRefId":     "subop_01HZ_NJERI_UPGRADE"
}

→ 200 OK
{
  "fulfillmentId":          "flb_01HZ_NJERI_UPGRADE",
  "terminalState":          "FULLY_PROVISIONED",
  "diffSummary": {
    "addedServices": [ "svc_IPTV_BASIC" ],
    "droppedServices": [],
    "modifiedServices": []
  },
  "serviceOutcomes": [
    { "serviceRef": "svc_IPTV_BASIC", "dispatchedVerb": "activateService", "outcome": "SUCCEEDED", "targetNodeCode": "OLT-NRB-WTL-01", "targetPort": "0/2/03", "adapterRef": "adp_01HZ_KARN_022" }
  ],
  "durationMs":              3500,
  "completedAt":             "2026-05-20T10:35:42+03:00"
}
```


5.3 Resume service (post-pause)

Called by RESUME-01's ful-call-activate worker (despite the activate name, the intent here is RESUME_SERVICE).

```
POST /api/service-fulfillment/execute HTTP/1.1
```

```
{
  "operatorCode":      "WIK",
  "subscriptionId":    "sub_01HZ_KIBE_FIBER",
  "customerId":        "cust_01HZ_KIBE",
  "packageRef":        "pkg_01HX_INET_50M",
  "packageVersionId":  "pkv_01HX_INET_50M_V2",
  "homepassId":        "hp_01HZ_NGNG_044",
  "intent":            "RESUME_SERVICE",
  "fulfillmentCorrelationId": "subop_01HZ_KIBE_RESUME:resume",
  "callerWorkflowKind": "SUB_WF_RESUME",
  "callerWorkflowRefId": "subop_01HZ_KIBE_RESUME"
}

→ 200 OK
{
  "fulfillmentId":  "flb_01HZ_KIBE_RESUME",
  "terminalState":  "FULLY_PROVISIONED",
  "serviceOutcomes": [
    { "serviceRef": "svc_INET_50M", "dispatchedVerb": "resumeService", "outcome": "SUCCEEDED", "targetNodeCode": "OLT-NRB-NGNG-02" },
  ],
  "durationMs":     1800
}
```

5.4 Errors

Missing equipment:

```
→ 200 OK
{
  "terminalState":  "FAILED_AT_FIRST_SERVICE",
  "failureReason":  "MISSING_REQUIRED_EQUIPMENT",
  "serviceOutcomes": [{ "serviceRef": "svc_INET_100M", "outcome": "FAILED", "failureDetail": "Required equipment for serviceClass" } ]
}
```

HomePass port not captured (AT_INSTALLATION mode without install completion):

```
→ 200 OK
{
  "terminalState":  "FAILED_AT_FIRST_SERVICE",
  "failureReason":  "HOMEPASS_PORT_NOT_CAPTURED",
  "serviceOutcomes": [{ "serviceRef": "svc_VOIP_STD", "outcome": "FAILED", "failureDetail": "HomePass hp_01HX_... is in AT_INSTALLATION mode" } ]
}
```

Partial provision with rollback:

```
→ 200 OK
{
  "terminalState":  "FAILED_AND_ROLLED_BACK",
  "failureReason":  "ADAPTER_TIMEOUT",
  "serviceOutcomes": [
    { "serviceRef": "svc_INET_100M", "outcome": "COMPENSATED", "durationMs": 1100 },
    { "serviceRef": "svc_IPTV_BASIC", "outcome": "FAILED", "failureDetail": "Verimatrix middleware timeout after 25s", "durationMs": 25000 }
  ]
}
```

(INET succeeded, then IPTV failed; INET was rolled back via deactivateService.)

6. Kafka events

Sample ServiceFulfillmentCompleted for Otieno's migration target activation:

```
{
  "eventType":  "ServiceFulfillmentCompleted",
  "aggregateId": "sub_01HZ_OTIENO_FIBER",
  "timestamp":  "2026-06-01T11:08:10+03:00",
  "payload": {
    "fulfillmentId":  "flb_01HZ_OTIENO_MIGRATE_ACTIVATE",
    "operatorCode":   "WIK",
    "subscriptionId": "sub_01HZ_OTIENO_FIBER",
    "customerId":     "cust_01HZ_OTIENO",
    "intent":         "ACTIVATE_SERVICE",
    "packageRef":     "pkg_01HX_INET_VOIP_IPTV_RES",
    "homepassId":     "hp_01HZ_LAVI_073",
    "terminalState":  "FULLY_PROVISIONED",
  }
}
```



```

    "serviceOutcomes": [
      {"serviceRef": "svc_INET_100M", "outcome": "SUCCEEDED"},
      {"serviceRef": "svc_VOIP_STD", "outcome": "SUCCEEDED"},
      {"serviceRef": "svc_IPTV_BASIC", "outcome": "SUCCEEDED"}
    ],
    "fulfillmentCorrelationId": "subop_01HZ_0TIEN0_MIGRATE:activate-target",
    "callerWorkflowKind": "SUB_WF_MIGRATION",
    "callerWorkflowRefId": "subop_01HZ_0TIEN0_MIGRATE",
    "completedAt": "2026-06-01T11:08:10+03:00"
  }
}

```

Sample ServiceFulfillmentFailed for an UPGRADE that hit IPTV adapter timeout:

```

{
  "eventType": "ServiceFulfillmentFailed",
  "aggregateId": "sub_01HZ_NJERI_FIBER",
  "timestamp": "2026-05-20T10:35:35+03:00",
  "payload": {
    "fulfillmentId": "flb_01HZ_NJERI_UPGRADE_TIMEOUT",
    "operatorCode": "WIK",
    "subscriptionId": "sub_01HZ_NJERI_FIBER",
    "intent": "MODIFY_SERVICE",
    "packageRef": "pkg_01HX_INET_100M_VOIP_RES",
    "targetPackageRef": "pkg_01HX_INET_VOIP_IPTV_RES",
    "homepassId": "hp_01HZ_KARN_022",
    "terminalState": "FAILED_AT_FIRST_SERVICE",
    "failureReason": "ADAPTER_TIMEOUT",
    "serviceOutcomes": [
      {"serviceRef": "svc_IPTV_BASIC", "outcome": "FAILED", "dispatchedVerb": "activateService", "failureDetail": "Verimatrix mic
    ],
    "fulfillmentCorrelationId": "subop_01HZ_NJERI_UPGRADE:modify",
    "failedAt": "2026-05-20T10:35:35+03:00"
  }
}

```

7. Cross-DD coordination

Module	Direction	Contract
SUB-WF-ACTIVATE-01	Caller	ful-call-activate worker; intent=ACTIVATE_SERVICE
SUB-WF-RESUME-01	Caller	ful-call-activate worker; intent=RESUME_SERVICE
SUB-WF-UPGRADE-01 / DOWNGRADE-01	Caller	ful-call-modify worker; intent=MODIFY_SERVICE with target Package
SUB-WF-RELOCATION-01	Caller	ful-call-activate worker; intent=ACTIVATE_SERVICE at target HomePass
SUB-WF-MIGRATION-01	Caller	ful-call-activate worker; intent=ACTIVATE_SERVICE at target HomePass with target Package composition
SIP-01	Reads Package + Package Version	Service composition, version pinning
PLM-CFG-01	Reads Service definitions + Provisioner interface	service_management_key, provisioner_key, network_profile_shape, is_addressable, consumption_model, equipment_requirement
RLM-CFG-01	Reads HomePass	service_management_endpoints, status_code
FUL-04	Sibling	Owns SUSPEND_SERVICE / APPLY_RESTRICTION / LIFT_RESTRICTION. FUL-04's resumeService is the compensation-only inverse of FUL-04's suspendService. FUL-03's resumeService is the legitimate post-pause restoration entry point.
FUL-05	Sibling	Owns network revocation on terminate.
FUL-07	Sibling	Owns HomePass-side bookkeeping (CLAIM / RELEASE). FUL-03 does not call FUL-07; the workflow calls both.
Provisioner adapters	Implementer	Implement the Provisioner interface (5 methods: activateService / modifyService / deactivateService / resumeService / suspendService). Registered per provisionerKey per PLM-CFG-01.

Module	Direction	Contract
NOT-01	Downstream consumer	Customer notification on fulfillment outcome.
NOC alerting	Downstream consumer	Compensation failures + adapter persistent errors.

7.1 Build order

This DD (v1.1) deploys after:

1. FUL-03 v1.0 (already deployed; v1.1 is a backwards-compatible extension)
2. PLM-CFG-01 v1.0 (declares the 5-method Provisioner interface; pending)
3. SIP-01 v1.0 (Package + Package Version pinning)
4. RLM-CFG-01 v1.0 (HomePass + service_management_endpoints)

Callers depending on v1.1 verbs:

- UPGRADE-01 / DOWNGRADE-01 (require MODIFY_SERVICE)
- RELOCATION-01 / MIGRATION-01 (use ACTIVATE_SERVICE against target HomePass — already in v1.0, no change needed for them)
- RESUME-01 (requires RESUME_SERVICE)

7.2 Stub debt + open coordination items

- **PLM-CFG-01 v1.0 Provisioner interface declaration** — pending. The 5-method interface (activateService / modifyService / deactivateService / resumeService / suspendService) needs to be formally specified in PLM-CFG-01 v1.0 before adapter implementations can target it.
- **v1.0 endpoint deprecation** — POST /api/activation/execute remains as an alias for intent=ACTIVATE_SERVICE. Marked deprecated; removal planned for v2.0. Existing ACTIVATE-01 callers continue without code change.
- **Equipment-binding validation** — FUL-03 does not validate that equipmentBindings[] covers all equipment_requirement Services. The caller (UPGRADE-01 / DOWNGRADE-01 / ACTIVATE-01 / RELOCATION-01 / MIGRATION-01) is responsible. If FUL-03 needs to enforce this in a future version, it would require reading PLM-CFG-01's equipment_requirement field per Service and rejecting on mismatch.

B — Current TZ

Status: To be filled in after codebase cartography review.

Legacy fulfillment-orchestration code exists in the TZ codebase (the v1.0 baseline was authored against partial TZ observation). Specific implementation details — whether the existing code does per-service-class fan-out, how it handles modify-service diff, whether it has separate methods for activate vs resume — require source-code review to document accurately. Until cartography lands, this DD does not assert specific legacy structural claims.